

Mental Health Text Classification — Literature Tables

Table 1: 2026 Papers

Title	Author(s)	Dataset	Best Model	F1 Score	Accuracy	Year
A contextual hierarchical attention network for detecting mental health disorders using social media	Ron Hochstenbach, Flavius Frasincar, Jasmijn Klinkhamer	eRisk (Self-Harm, Anorexia, Depression)	Context-HAN	0.69 (Depression)	Not Reported	2026
A large-scale annotated Urdu corpus and deep learning benchmark for mental health classification	Maleeka Fatima, Muhammad Saleem Khan, Muhammad Shahzad Faisal, et al.	Urdu tweets (36,000)	UrduBERT	0.8161 (Macro)	81.71%	2026
Early detection of mental health on social media using a hybrid Bi-LSTM-XGBoost model	Heru Syah Putra, Muhammad Ary Murti, Astri Novianty, Nur Hafieza Ismaili	Kaggle Mental Health Dataset (42,029)	Bi-LSTM-XGBoost	0.90 (Weighted)	90.35%	2026
Deep-AP: An efficient multi-task architecture for Author Profiling and Depression Detection	Idriss Oulahbib, Meriem Benhaddi, Salah El Hadaï	Reddit (2005–2023)	Deep-AP BERT-large	0.881 (Depression)	95.8%	2026
A comparative and statistical analysis of depression classification in social media	Miryam Elizabeth Villa-Pérez, Karla María Valencia-Segura, Daniela Moctezuma, et al.	Spanish/English Twitter, CLPsych2015	TF-IDF + Emotions	0.82 (Spanish)	Not Reported	2026
Cross platform social media analysis for mental health detection	Kanishka Bhatt, Ashwini Kumar Singh, Priyank Pandey, et al.	Multiple (Reddit, Twitter, etc.)	RoBERTa	>0.90	Not Reported	2026
A unified comparative evaluation of machine learning, deep learning and GPT-2 for suicide ideation detection	Yasmeen Mohamed Saleh, Fahad Kamal Alsheref, Mahmoud Mohamed Bahloul	Reddit (SuicideWatch)	GPT-2 (Fine-tuned)	0.9825	98.25%	2026
Enhancing depression classification from social media using transformer-based ensemble models and DSM-5 criteria	Suksam J., Doenribram D., Nuankaew W.S., Nuankaew P., Chaisoong U., Jareanpon C.	Twitter (9 DSM-5 symptoms)	Bagging	0.9394 (Symptom)	93.98%	2026

Table 2: 2025 Papers

Title	Author(s)	Dataset	Best Model	F1 Score	Accuracy	Year
Comparing traditional NLP and large language models for mental health status classification	Thomas Kallstenius, Andrea Johansson Capusan, Gerhard Andersson, Adam Williamson	Social media (52,681 statements)	NLP + SVM (RBF)	0.97 (Normal)	95%	2025
Detecting Mental Health Conditions from Reddit Posts Using Machine Learning	Mujeeb Ur Rehman, Menahil Khawar, Sohail Khalid, et al.	Reddit (53,043 posts)	Stacking Ensemble	0.92 (Macro)	91%	2025
Multi task opinion enhanced hybrid BERT model for mental health analysis	Md. Mithun Hossain, Md. Shakil Hossain, M. F. Mridha, et al.	Multiple mental health datasets	Opinion-BERT	0.9432 (Status)	93.74% (Status)	2025
Trade-offs between machine learning and deep learning for mental illness detection on social media	Zhanyi Ding, Zhongyan Wang, Yeyubei Zhang, et al.	Kaggle Mental Health Dataset	ALBERT	0.9650 (Binary)	Not Reported	2025
A soft voting ensemble model for suicide ideation detection	Erol KINA, Jin-Ghoo Choi, Abid Ishaq, et al.	Reddit SuicideWatch + Depression (232,074)	SVEM (Soft Voting Ensemble)	0.94	94%	2025
Biomimetic modeling for mental health: MCoG-LDPSNet	Christos Bormpotsis, M. Anagnostou, M. Sedky, E. Jastrzebska, A. Patel	Kaggle Mental Health Corpus	MCoG-LDPSNet	0.9451 (G-mean)	96.60%	2025
Analyzing student mental health with RoBERTa-Large	Hikmat Ullah Khan, Anam Naz, Fawaz Khaled Alarfaj, Naif Almusallam	Student mental health dataset	RoBERTa-Large	0.94	97%	2025

Table 3: Mental Health Text Classification — Total Papers Summary (All 15 Papers)

#	Title	Author(s)	Dataset	Best Model	F1 Score	Accuracy	Year
1	A contextual hierarchical attention network for detecting mental health disorders using social media	Ron Hochstenbach, Flavius Frasincar, Jasmijn Klinkhamer	eRisk (Self-Harm, Anorexia, Depression)	Context-HAN	0.69 (Depression)	—	2026
2	A large-scale annotated Urdu corpus and deep learning benchmark for mental health classification	Maleeka Fatima, Muhammad Saleem Khan, Muhammad Shahzad Faisal, et al.	Urdu tweets (36,000)	UrduBERT	0.8161 (Macro)	81.71%	2026
3	Early detection of mental health on social media using a hybrid Bi-LSTM-XGBoost model	Heru Syah Putra, Muhammad Ary Murti, Astri Novianty, Nur Hafieza Ismaili	Kaggle Mental Health Dataset (42,029)	Bi-LSTM-XGBoost	0.90 (Weighted)	90.35%	2026
4	Deep-AP: An efficient multi-task architecture for Author Profiling and Depression Detection	Idriss Oulahbib, Meriem Benhaddi, Salah El Hadaj	Reddit (2005–2023)	Deep-AP BERT-large	0.881 (Depression)	95.8%	2026
5	A comparative and statistical analysis of depression classification in social media	Miryam Elizabeth Villa-Pérez, Karla María Valencia-Segura, Daniela Moctezuma, et al.	Spanish/English Twitter, CLPsych2015	TF-IDF + Emotions	0.82 (Spanish)	—	2026
6	Cross platform social media analysis for mental health detection	Kanishka Bhatt, Ashwini Kumar Singh, Priyank Pandey, et al.	Multiple (Reddit, Twitter, etc.)	RoBERTa	>0.90	—	2026
7	A unified comparative evaluation of machine learning, deep learning and GPT-2 for suicide ideation detection	Yasmeen Mohamed Saleh, Fahad Kamal Alsheref, Mahmoud Mohamed Bahloul	Reddit (SuicideWatch)	GPT-2 (Fine-tuned)	0.9825	98.25%	2026
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14	Analyzing student mental health with RoBERTa-Large	Hikmat Ullah Khan, Anam Naz, Fawaz Khaled Alarfaj, Naif Almusallam	Student mental health dataset	RoBERTa-Large	0.94	97%	2025
15	Enhancing depression classification from social media using transformer-based ensemble models and DSM-5 criteria	Suksam J., Doenribram D., Nuankaew W.S., Nuankaew P., Chaisoong U., Jareanpon C.	Twitter (9 DSM-5 symptoms)	Bagging	0.9394 (Symptom)	93.98%	2026

Note: '—' indicates the original paper did not report accuracy.